





Generative Models. VAE

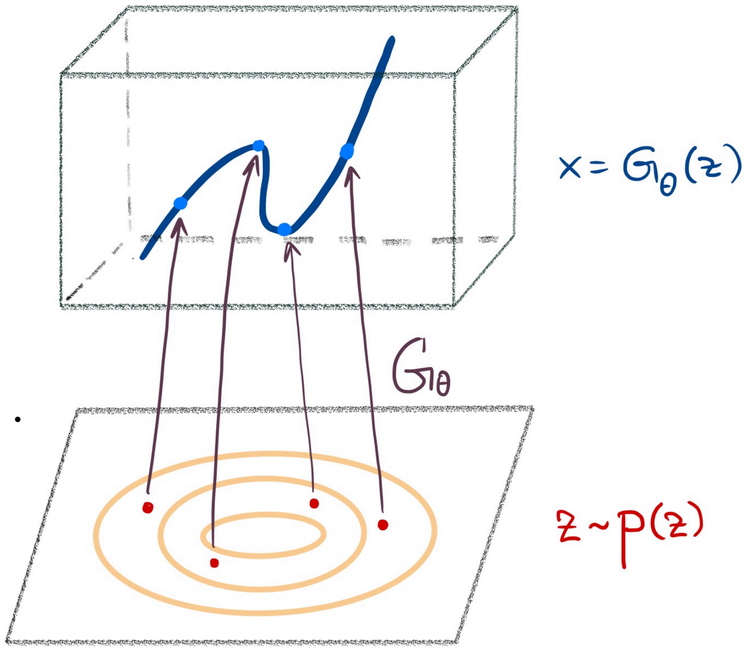
Denis Rakitin

Generative Modeling: problem

Given $\mathbf{x}_1, \ldots, \mathbf{x}_n \sim p_{\text{data}}(\mathbf{x})$, generate new samples $\hat{\mathbf{x}}_1, \ldots, \hat{\mathbf{x}}_n \sim p_{\text{data}}(\mathbf{x})$

- $\mathbf{x} \in \mathbb{R}^D$, e.g. images (then $D = H \times W \times C$);
- Typical way: introduce a neural network generator $G_\theta(\mathbf{z})$;
- Latent variable $\mathbf{z} \sim p(\mathbf{z})$ should come from an easy distribution

  e.g. $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$;
- Generator should produce distribution $p_\theta(\mathbf{x})$, close to the data: $p_\theta \approx p_{\text{data}}$.

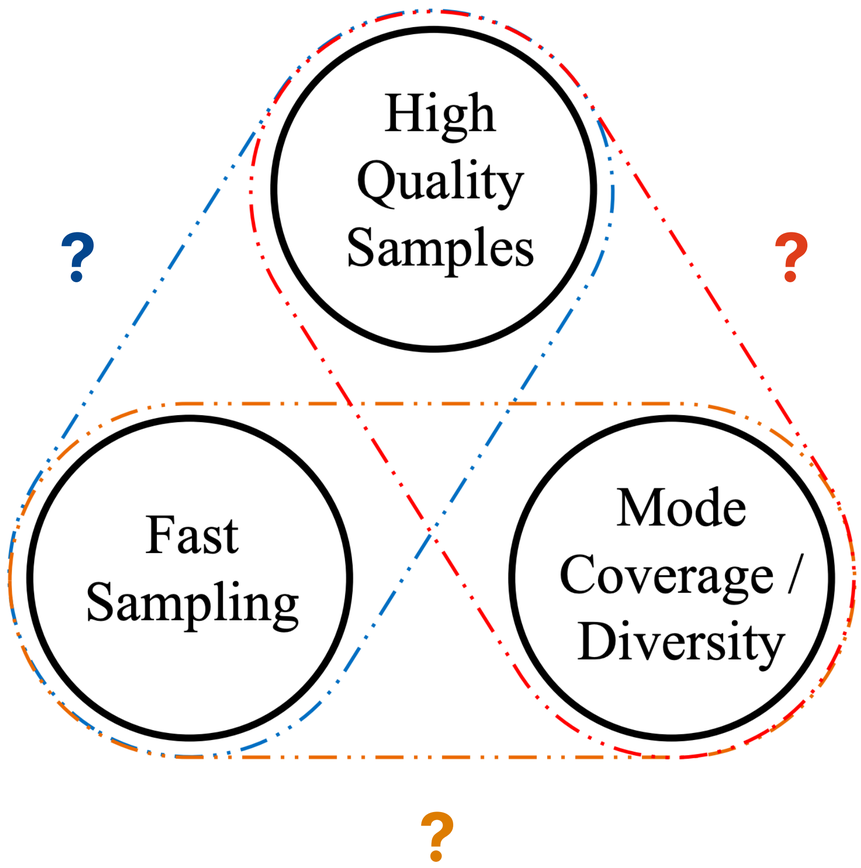


All generative models' training follows the form

$$d(p_\theta; p_{\text{data}}) \longrightarrow \min_\theta$$

for some distance function d.

Generative Trilemma

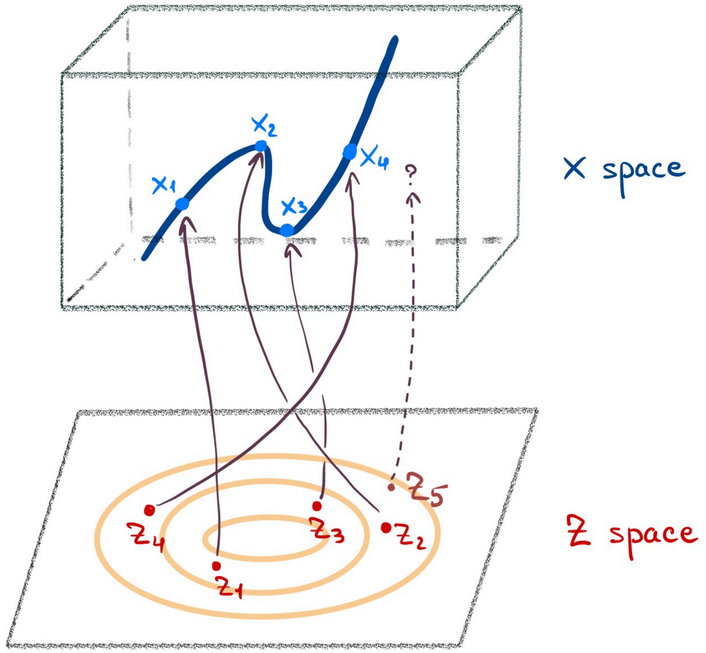


Z. Xiao et al. Tackling the Generative Learning Trilemma with Denoising Diffusion GANs

Naive approach

For $\mathbf{z} \sim p(\mathbf{z})$ we want $G_\theta(\mathbf{z})$ to be similar to $\mathbf{x} \sim p_{\text{data}}(\mathbf{x})$.

- What if we encourage it directly, e.g. by minimizing $\frac{1}{n}\sum_{i=1}^{n} \|G_\theta(\mathbf{z}_i) - \mathbf{x}_i\|^2$?
- We can take $\mathbf{x}_1, \ldots, \mathbf{x}_n \sim p_{\text{data}}(\mathbf{x})$, but what are $\mathbf{z}_1, \ldots, \mathbf{z}_n$?
- A possible choice is generating them from $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$.



The resulting procedure is unlikely to perform well:

- The generator learns, essentially, a random transformation;
- It can be arbitrarily complex → hard to generalize to new samples.

Autoencoders

AEs are built on the same idea of minimizing $\frac{1}{n}\sum_{i=1}^{n} \|G_\theta(\mathbf{z}_i) - \mathbf{x}_i\|^2$.

- Random $\mathbf{z}_i$ are bad. How to choose them appropriately?
- Automatically! Let a neural network learn it.

Let $E_\varphi(\mathbf{x})$ be this *encoder* network which predicts the latent samples.

Training objective:

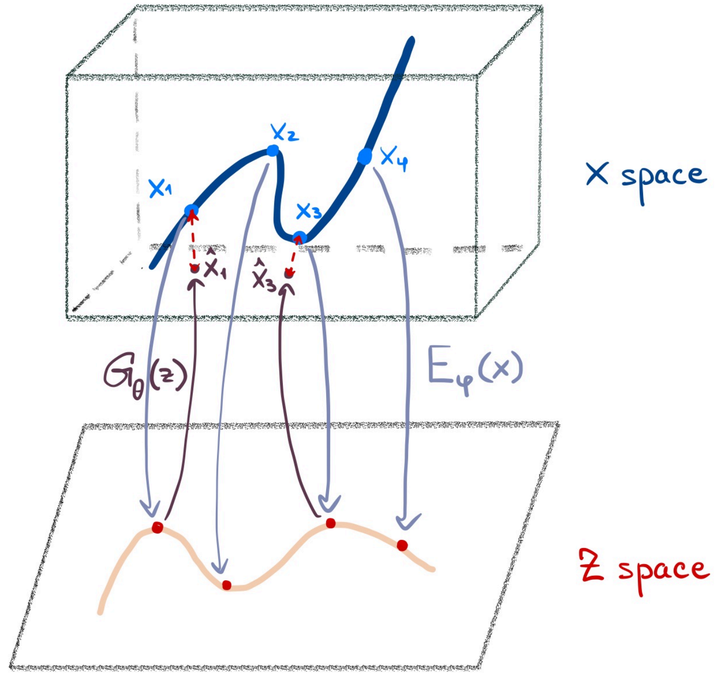
$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| G_\theta\big(E_\varphi(\mathbf{x})\big) - \mathbf{x} \right\|^2 \longrightarrow \min_{\varphi,\theta}$$

Autoencoders (AEs): properties

Training objective:

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \left\| G_\theta\big(E_\varphi(\mathbf{x})\big) - \mathbf{x} \right\|^2 \longrightarrow \min_{\varphi,\theta}$$

- $G_\theta(\mathbf{z})$, the *decoder* network, should be inverse to the encoder;
- The latent space is typically lower-dimensional;
- The encoder performs compression.



How to generate at inference time?

- The distribution of $E_\varphi(\mathbf{x})$ is unknown → need to learn it.

Can reduce the dimensionality of the problem, but does not solve it.

Road to VAEs

- Autoencoders incorporate only the functional equivalence;
- There is no distributional constraints → encoder produces arbitrary latent distribution.

$$\mathbf{x} \sim p_{\text{data}}(\mathbf{x}) \qquad \mathbf{z} = E_\varphi(\mathbf{x})$$

$$\downarrow\uparrow$$

$$\hat{\mathbf{x}} = G_\theta\big(E_\varphi(\mathbf{x})\big) \quad \mathbf{z} = E_\varphi(\mathbf{x})$$

- Variational Autoencoders consider two probabilistic models: encoding real data, and decoding Gaussian latents;
- VAEs maximizes similarity between these models;
- Encoder should produce Gaussian latents;
- Decoder should produce realistic data samples.

$$\mathbf{x} \sim p_{\text{data}}(\mathbf{x}) \qquad \mathbf{z} = E_\varphi(\mathbf{x})$$

$$\downarrow\uparrow$$

$$\hat{\mathbf{x}} = G_\theta(\hat{\mathbf{z}}) \qquad \hat{\mathbf{z}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Variational Autoencoders

VAEs introduce two probabilistic models. Data generation / decoding:

$$p_\theta(\mathbf{x}, \mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})\, p_\theta(\mathbf{x} \mid \mathbf{z}); \;\; p_\theta(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}\big(\mathbf{x} \mid G_\theta(\mathbf{z}), \sigma^2 \mathbf{I}\big).$$

- Latent sample is standard Gaussian;
- The decoded data sample is obtained as $\mathbf{x} = G_\theta(\mathbf{z}) + \sigma\varepsilon$.

Data encoding:

$$q_\varphi(\mathbf{x}, \mathbf{z}) = p_{\text{data}}(\mathbf{x})\, q_\varphi(\mathbf{z} \mid \mathbf{x}); \quad q_\varphi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}\big(\mathbf{z} \mid E_\varphi(\mathbf{x}), \sigma_\varphi^2 \mathbf{I}\big).$$

- The $\mathbf{x}$ sample comes from real data;
- The encoded sample is obtained as $\mathbf{z} = E_\varphi(\mathbf{x}) + \sigma_\varphi \cdot \varepsilon$.

D. Kingma, M. Welling. Auto-Encoding Variational Bayes

Variational Autoencoders

Data generation / decoding:

$$p_\theta(\mathbf{x}, \mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})\, p_\theta(\mathbf{x} \mid \mathbf{z}); \quad p_\theta(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}\big(\mathbf{x} \mid G_\theta(\mathbf{z}), \sigma^2 \mathbf{I}\big).$$

Data encoding:

$$q_\varphi(\mathbf{x}, \mathbf{z}) = p_{\text{data}}(\mathbf{x})\, q_\varphi(\mathbf{z} \mid \mathbf{x}); \quad q_\varphi(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}\big(\mathbf{z} \mid E_\varphi(\mathbf{x}), \sigma_\varphi^2 \mathbf{I}\big).$$

Goal: the equivalence of two models. Optimization problem:

$$\mathrm{KL}(q_\varphi(\mathbf{x}, \mathbf{z}) \| p_\theta(\mathbf{x}, \mathbf{z})) \to \min_{\theta, \varphi}$$

VAEs: derivation

Derivation of the optimization procedure:

$$\mathrm{KL}(q_\varphi(\mathbf{x},\mathbf{z}) \,\|\, p_\theta(\mathbf{x},\mathbf{z})) = \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_\varphi(\mathbf{z}|\mathbf{x})} \log \frac{p_{\text{data}}(\mathbf{x})q_\varphi(\mathbf{z} \mid \mathbf{x})}{\mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})\, p_\theta(\mathbf{x} \mid \mathbf{z})}$$

$$= \underbrace{\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \log p_{\text{data}}(\mathbf{x})}_{\text{const}} + \mathbb{E}_{p_{\text{data}}(\mathbf{x})}\mathbb{E}_{q_\varphi(\mathbf{z}|\mathbf{x})} \log \frac{q_\varphi(\mathbf{z} \mid \mathbf{x})}{p(\mathbf{z})} - \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_\varphi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} \mid \mathbf{z})$$

$$= \text{const} + \boxed{\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathrm{KL}(q_\varphi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z}))} - \mathbb{E}_{p_{\text{data}}(\mathbf{x})q_\varphi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} \mid \mathbf{z}).$$

KL between Gaussians, computationally tractable

VAEs: derivation

Recall the Gaussian decoder likelihood:

$$p_\theta(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}\big(\mathbf{x} \mid G_\theta(\mathbf{z}), \sigma^2 \mathbf{I}\big) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{x} - G_\theta(\mathbf{z})\|^2\right).$$

The loss is equivalent to

$$\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathrm{KL}(q_\varphi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})) - \mathbb{E}_{p_{\text{data}}(\mathbf{x}) q_\varphi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x} \mid \mathbf{z})$$

$$= \text{const} + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathrm{KL}(q_\varphi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})) + \mathbb{E}_{p_{\text{data}}(\mathbf{x}) q_\varphi(\mathbf{z}|\mathbf{x})} \frac{1}{2\sigma^2} \|\mathbf{x} - G_\theta(\mathbf{z})\|^2$$

$$= \text{const} + \boxed{\mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathrm{KL}(q_\varphi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z}))} + \boxed{\mathbb{E}_{p_{\text{data}}(\mathbf{x}) \mathcal{N}(\varepsilon|\mathbf{0},\mathbf{I})} \frac{1}{2\sigma^2} \left\|\mathbf{x} - G_\theta\big(E_\varphi(\mathbf{x}) + \sigma_\varphi \varepsilon\big)\right\|^2}.$$

promoting Gaussian latents

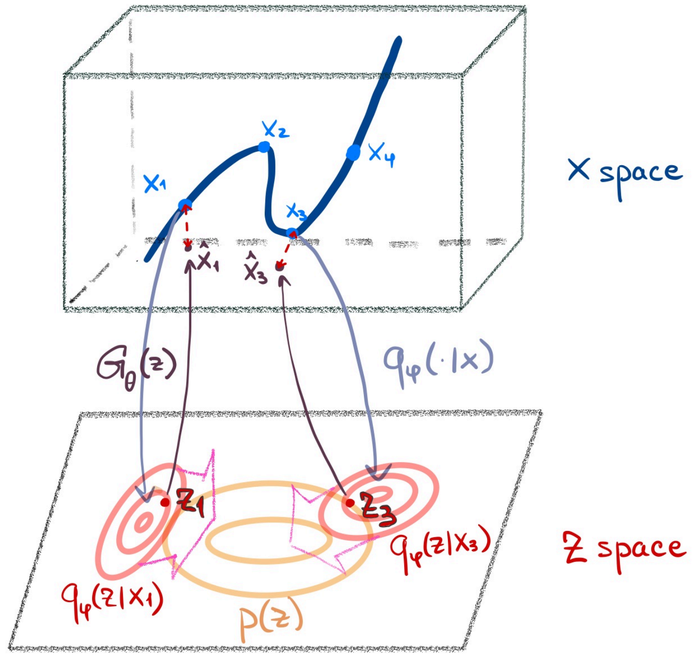
reconstruction loss (almost as in AEs)

VAEs: summary

$$\mathrm{KL}(q_\varphi(\mathbf{x},\mathbf{z}) \,\|\, p_\theta(\mathbf{x},\mathbf{z})) =$$

$$= \text{const} + \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathrm{KL}(q_\varphi(\mathbf{z} \mid \mathbf{x}) \,\|\, p(\mathbf{z})) +$$

$$+ \mathbb{E}_{p_{\text{data}}(\mathbf{x})\mathcal{N}(\varepsilon|\mathbf{0},\mathbf{I})} \frac{1}{2\sigma^2} \left\| \mathbf{x} - G_\theta\big(E_\varphi(\mathbf{x}) + \sigma_\varphi \varepsilon\big) \right\|^2.$$



Variational Autoencoders =

- Autoencoders' reconstruction loss +
- Stochastic encoder +
- KL between latents and standard Gaussian distribution;
- The latter allows to successfully decode Gaussian latents at inference.

VAEs: theory

In theory, global minimum implies equivalence of the two models:

$$\mathrm{KL}(q_\varphi(\mathbf{x},\mathbf{z}) \,\|\, p_\theta(\mathbf{x},\mathbf{z})) = 0 \quad \Longrightarrow \quad q_\varphi(\mathbf{x},\mathbf{z}) = p_\theta(\mathbf{x},\mathbf{z})$$

Equal joint distributions imply equal marginals:

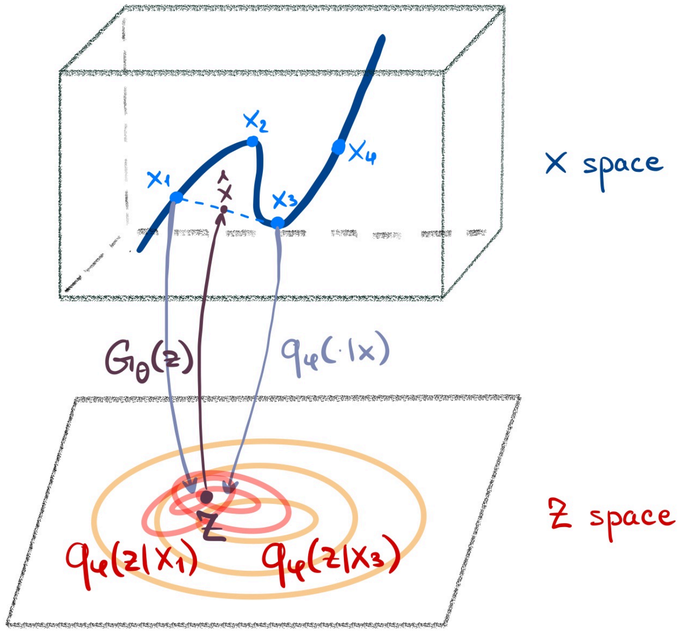
$$q_\varphi(\mathbf{x},\mathbf{z}) = p_\theta(\mathbf{x},\mathbf{z}) \quad \Longrightarrow \quad \begin{cases} q_\varphi(\mathbf{z}) = p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I}), \\ p_{\mathrm{data}}(\mathbf{x}) = p_\theta(\mathbf{x}). \end{cases}$$

- Optimal encoder produces Gaussian latents;
- Optimal decoder produces samples from the data distribution;
- Decoding Gaussian latents at inference is theoretically justified.

VAEs: practice

Where is VAE in the Generative Trilemma?

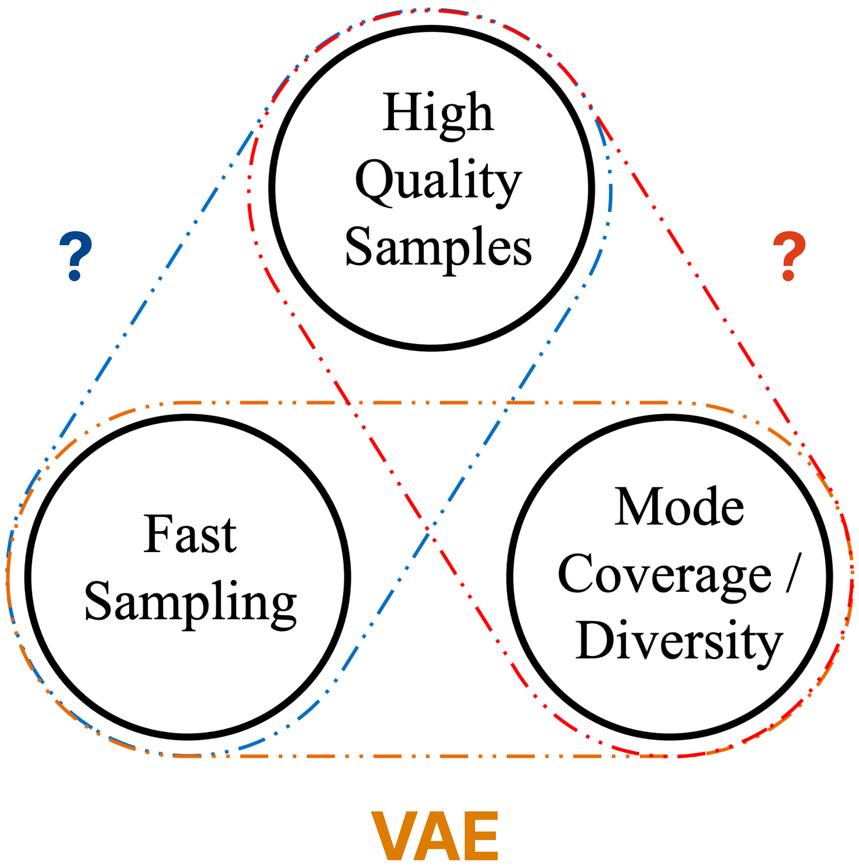
- VAE generation is fast: one decoder network inference;
- Decoder should cover all data modes by design (via reconstruction of the dataset);
- Individual samples, however, are not always good.



The **encoder's** problem is hard. It is typically the source of quality degradation:

- If encoder merges different latents, decoder performs averaging;
- Good decoder + encoder, unable to produce Gaussian distribution = poor samples at generation (same problem as in AEs).

VAEs: Generative Trilemma



VAEs: conclusion

- Main idea: matching two probabilistic models instead of functions;
- Fast generation;
- Good mode coverage;
- Applicable for dimensionality reduction;
- Insufficient quality of individual samples.

The approach is rich in ideas and fundamental in generative modeling.

Studying it is helpful for newer models:

- Diffusion models can be seen as VAEs with fixed but very powerful encoder.